

Supplementary Material

for Active Momentum Control for In-Pipe Robots in Microgravity: On-Orbit Validation

SI. GENERAL NOTATION AND SYMBOL CONVENTIONS

To ensure strict clarity and adhere to standard academic typography, the following mathematical notation conventions are consistently applied throughout the main manuscript and this supplementary material:

- **Vectors and Matrices** are denoted by bold, non-italicized letters (e.g., $\mathbf{q}$, $\mathbf{A}$, $\boldsymbol{\nu}$).
- **Scalars and Constants** are denoted by standard italicized letters (e.g., m_i , N , α_{decay}).
- **Coordinate Frames and Sets** are denoted by calligraphic letters (e.g., $\mathcal{W}$, $\mathcal{B}$, $\mathcal{M}$).

A comprehensive nomenclature and system variable definitions are detailed in Table S1.

TABLE S1: Nomenclature and System Variables

Symbol	Dim.	Definition
Coordinate Systems		
$\mathcal{W}$	-	Inertial/World frame, fixed to a reference point on the pipe.
$\mathcal{B}$	-	Base frame, attached to the geometric center of the link 1.
State Variables		
$\mathbf{q}$	$\mathbb{R}^{16}$	Generalized coordinates, defined on $\mathcal{Q} = \mathbb{R}^3 \times SO(3) \times \mathbb{R}^5 \times \mathbb{R}^4$ (excluding wheel angles).
$\boldsymbol{\nu}$	$\mathbb{R}^{27}$	Generalized velocities in tangent space, including wheel rotational velocities $\dot{\mathbf{q}}_w$.
$\mathbf{p}_b$	$\mathbb{R}^3$	Base position, coordinates of the geometric center in $\mathcal{W}$.
$\boldsymbol{\xi}_b$	$\mathbb{R}^4$	Base orientation, represented as a unit quaternion (Hamiltonian) in $\mathcal{W}$.
$\delta\phi_b$	$\mathbb{R}^3$	Attitude error, represented as a Lie algebra/rotation vector (optimization variable in QP).
$\mathbf{q}_s/\mathbf{q}_l/\mathbf{q}_w$	$\mathbb{R}^5 / \mathbb{R}^4 / \mathbb{R}^{12}$	Joint angles (spine), extension lengths (legs), and rotation angles (wheels).
Dynamics		
$\mathbf{h}_{com}$	$\mathbb{R}^6$	Centroidal momentum, defined as $[\mathbf{k}^\top (\text{angular}), \mathbf{I}^\top (\text{linear})]^\top$.
Environment		
$\hat{\mu}$	$\mathbb{R}^1$	Estimated friction coefficient, a real-time updated safety parameter.
Constraints		
f_{min}^{clamp}	$\mathbb{R}^1$	Minimum required normal force to maintain reliable contact.
F_{wall}^{press}	$\mathbb{R}^1$	Wall pressure limit, serving as a normal environmental hard constraint.
F_{wall}^{shear}	$\mathbb{R}^1$	Wall shear limit, serving as a tangential environmental hard constraint.

SII. DERIVATION OF THE CENTROIDAL MOMENTUM MATRIX (CMM)

This supplementary section provides the rigorous mathematical derivation of the Centroidal Momentum Matrix (CMM), $\mathbf{A}(\mathbf{q})$, and its sub-matrices ($\mathbf{A}_b$, $\mathbf{A}_s$, $\mathbf{A}_l$, and $\mathbf{A}_w$) as utilized in the hierarchical control framework.

We employ spatial vector algebra to describe the kinematics and dynamics of the multi-body system. The robot consists of

TABLE S2: Summary of Key Control Parameters and Safety Thresholds

Parameter / Threshold	Final Value	Physical Justification
Wall Pressure (F_{wall}^{press})	6.6 N	CSS operational structural limit
Min. Clamping (f_{min}^{clamp})	> 0 N	Detachment prevention
Wall Shear (F_{wall}^{shear})	Confidential	Bounded by friction cone
Joint Torque (τ_{max})	0.45 Nm	Motor peak mechanical output
Peak Friction (μ_{max})	0.576	Maximum estimated traction
Friction Floor (μ_{min})	0.40	Adaptive clipping boundary
MPC Horizon (N)	10 steps	0.2s dynamic look-ahead
WBC Control Frequency	400 Hz	High-bandwidth safety margin

TABLE S3: WBC Computation Time Distribution (Cores 0-1)

Samples	Mean	Std. Dev.	Worst-Case
$> 20,000$	1.7 ms	0.15 ms	2.2 ms

N_b rigid bodies, including the base link, spinal links, telescopic leg segments, and active wheels.

A. System Center of Mass and Spatial Kinematics

Let m_i be the mass of the i -th rigid body, and $\mathbf{p}_i \in \mathbb{R}^3$ be the position of its local center of mass (CoM) in the inertial frame $\mathcal{W}$. The system's total mass is $M = \sum_{i=1}^{N_b} m_i$, and the system's global CoM position is given by:

$$\mathbf{p}_{com} = \frac{1}{M} \sum_{i=1}^{N_b} m_i \mathbf{p}_i \quad (\text{S1})$$

For each rigid body i , its spatial velocity $\mathbf{V}_i = [\boldsymbol{\omega}_i^\top, \mathbf{v}_i^\top]^\top \in \mathbb{R}^6$ expressed in its local frame $\mathcal{B}_i$ can be mapped from the system's generalized velocity $\boldsymbol{\nu} = [\mathbf{v}_b^\top, \boldsymbol{\omega}_b^\top, \dot{\mathbf{q}}_s^\top, \dot{\mathbf{q}}_l^\top, \dot{\mathbf{q}}_w^\top]^\top$ via its body Jacobian $\mathbf{J}_i(\mathbf{q})$:

$$\mathbf{V}_i = \mathbf{J}_i(\mathbf{q}) \boldsymbol{\nu} \quad (\text{S2})$$

B. Spatial Momentum of a Single Rigid Body

The spatial inertia matrix of the i -th rigid body at its local CoM is denoted as $\mathbf{I}_i \in \mathbb{R}^{6 \times 6}$, defined as:

$$\mathbf{I}_i = \begin{bmatrix} \bar{\mathbf{I}}_i & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & m_i \mathbf{1}_{3 \times 3} \end{bmatrix} \quad (\text{S3})$$

where $\bar{\mathbf{I}}_i \in \mathbb{R}^{3 \times 3}$ is the rotational inertia tensor, and $\mathbf{1}_{3 \times 3}$ is the identity matrix. The spatial momentum of body i , expressed in its local frame, is:

$$\mathbf{h}_i = \mathbf{I}_i \mathbf{V}_i = \mathbf{I}_i \mathbf{J}_i(\mathbf{q}) \boldsymbol{\nu} \quad (\text{S4})$$

C. Adjoint Transformation to the System CoM

To calculate the total centroidal momentum $\mathbf{h}_{com} \in \mathbb{R}^6$, the local momentum $\mathbf{h}_i$ of each body must be translated to the system's global CoM frame $\mathcal{C}$. We define the spatial force/momentum adjoint transformation matrix from frame $\mathcal{B}_i$ to the CoM frame $\mathcal{C}$ as ${}^{com}\mathbf{X}_i^* \in \mathbb{R}^{6 \times 6}$:

$${}^{com}\mathbf{X}_i^* = \begin{bmatrix} {}^{com}\mathbf{R}_i & \mathbf{S}(\mathbf{p}_{com} - \mathbf{p}_i){}^{com}\mathbf{R}_i \\ \mathbf{0}_{3 \times 3} & {}^{com}\mathbf{R}_i \end{bmatrix} \quad (S5)$$

where ${}^{com}\mathbf{R}_i \in SO(3)$ is the rotation matrix from frame $\mathcal{B}_i$ to the inertial-aligned CoM frame, and $\mathbf{S}(\cdot)$ is the skew-symmetric cross-product matrix.

D. Formulation of the CMM

By the principle of superposition, the total centroidal momentum $\mathbf{h}_{com}$ is the sum of the transformed spatial momenta of all individual bodies:

$$\mathbf{h}_{com} = \sum_{i=1}^{N_b} {}^{com}\mathbf{X}_i^* \mathbf{h}_i = \left(\sum_{i=1}^{N_b} {}^{com}\mathbf{X}_i^* \mathbf{I}_i \mathbf{J}_i(\mathbf{q}) \right) \boldsymbol{\nu} \quad (S6)$$

By definition, the matrix mapping the generalized velocity $\boldsymbol{\nu}$ to the centroidal momentum $\mathbf{h}_{com}$ is the Centroidal Momentum Matrix $\mathbf{A}(\mathbf{q}) \in \mathbb{R}^{6 \times (6+n_s+n_l+n_w)}$:

$$\mathbf{A}(\mathbf{q}) = \sum_{i=1}^{N_b} {}^{com}\mathbf{X}_i^* \mathbf{I}_i \mathbf{J}_i(\mathbf{q}) \quad (S7)$$

E. Sub-Matrix Decomposition

Since the generalized velocity vector is explicitly partitioned according to the kinematic chains: $\boldsymbol{\nu} = [\boldsymbol{\nu}_b^\top, \dot{\mathbf{q}}_s^\top, \dot{\mathbf{q}}_l^\top, \dot{\mathbf{q}}_w^\top]^\top$, the body Jacobian $\mathbf{J}_i(\mathbf{q})$ can be correspondingly partitioned into block columns:

$$\mathbf{J}_i = [\mathbf{J}_{i,b} \quad \mathbf{J}_{i,s} \quad \mathbf{J}_{i,l} \quad \mathbf{J}_{i,w}] \quad (S8)$$

Substituting this partition into Eq. (S7), the CMM naturally decomposes into four sub-matrices:

$$\mathbf{A}(\mathbf{q}) = [\mathbf{A}_b \quad \mathbf{A}_s \quad \mathbf{A}_l \quad \mathbf{A}_w] \quad (S9)$$

where each sub-matrix represents the physical momentum coupling of a specific kinematic chain to the whole body:

$$\mathbf{A}_b = \sum_{i=1}^{N_b} {}^{com}\mathbf{X}_i^* \mathbf{I}_i \mathbf{J}_{i,b} \in \mathbb{R}^{6 \times 6} \quad (\text{Floating Base}) \quad (S10)$$

$$\mathbf{A}_s = \sum_{i=1}^{N_b} {}^{com}\mathbf{X}_i^* \mathbf{I}_i \mathbf{J}_{i,s} \in \mathbb{R}^{6 \times 5} \quad (\text{Spinal Joints}) \quad (S11)$$

$$\mathbf{A}_l = \sum_{i=1}^{N_b} {}^{com}\mathbf{X}_i^* \mathbf{I}_i \mathbf{J}_{i,l} \in \mathbb{R}^{6 \times 4} \quad (\text{Leg Extensions}) \quad (S12)$$

$$\mathbf{A}_w = \sum_{i=1}^{N_b} {}^{com}\mathbf{X}_i^* \mathbf{I}_i \mathbf{J}_{i,w} \in \mathbb{R}^{6 \times 12} \quad (\text{Active Wheels}) \quad (S13)$$

This formulation strictly proves that even if the base remains stationary ($\boldsymbol{\nu}_b = \mathbf{0}$) and external contacts are absent, the active modulation of the spinal joints ($\dot{\mathbf{q}}_s$) directly regulates the internal momentum flux $\mathbf{h}_{com}^{internal} = \mathbf{A}_s \dot{\mathbf{q}}_s$, thereby generating the required reaction torques for omnidirectional reorientation in microgravity.

SIII. LIE-ALGEBRAIC ERROR-STATE KINEMATICS AND MPC LINEARIZATION

This section provides the rigorous mathematical derivation of the discrete-time linearized Jacobians ($\mathbf{A}_k$ and $\mathbf{B}_k$) on the $SO(3)$ manifold, which are utilized in the Error-State Model Predictive Control (MPC) formulation (Eq. 16-17 in the main text).

A. Error-State Representation on the Manifold

The floating-base robot's state space evolves on a differentiable manifold $\mathcal{M} = \mathbb{R}^3 \times SO(3) \times \mathbb{R}^{n_j}$. To avoid singularities associated with Euler angles and maintain the unit-norm constraint of quaternions, we employ an error-state formulation.

Let the true state be $\mathbf{x}$, the nominal state evaluated by the nonlinear dynamics be $\bar{\mathbf{x}}$, and the error state in the local tangent space be $\delta\mathbf{x}$. The orientation error $\delta\phi_b \in \mathbb{R}^3$ is defined in the Lie algebra $\mathfrak{so}(3)$ mapped to the $SO(3)$ manifold via the exponential map:

$$\boldsymbol{\xi}_{true} = \bar{\boldsymbol{\xi}} \otimes \text{Exp}(\delta\phi_b) \quad (S14)$$

where $\otimes$ denotes quaternion multiplication, and $\text{Exp}(\cdot)$ maps a rotation vector to a unit quaternion. For small angular errors, $\text{Exp}(\delta\phi_b) \approx [1, \frac{1}{2}\delta\phi_b^\top]^\top$.

The full error state vector in the tangent space is defined as:

$$\delta\mathbf{x} = [\delta\mathbf{h}_{com}^\top, \delta\mathbf{p}_b^\top, \delta\phi_b^\top, \delta\mathbf{q}_s^\top]^\top \in \mathbb{R}^{6+3+3+5} \quad (S15)$$

B. Continuous-Time Error Dynamics

1) *Momentum Dynamics*: The centroidal momentum dynamics are governed by the external contact wrenches. The continuous-time error dynamics are obtained by applying the first-order Taylor expansion to the Newton-Euler equations:

$$\delta\dot{\mathbf{h}}_{com} = \sum_{i=1}^{n_c} \left(\frac{\partial \mathbf{W}_i}{\partial \mathbf{p}_b} \delta\mathbf{p}_b + \frac{\partial \mathbf{W}_i}{\partial \phi_b} \delta\phi_b \right) \bar{\mathbf{f}}_{c,i} + \mathbf{W}_i \delta\mathbf{f}_{c,i} \quad (S16)$$

where $\mathbf{W}_i$ is the contact wrench matrix.

2) *Attitude Kinematics on $SO(3)$* : The true angular velocity $\boldsymbol{\omega}_b$ and the nominal angular velocity $\bar{\boldsymbol{\omega}}_b$ satisfy the continuous-time attitude kinematics. Utilizing the Baker-Campbell-Hausdorff (BCH) approximation, the differential equation for the attitude error $\delta\phi_b$ is derived as:

$$\delta\dot{\phi}_b \approx \delta\boldsymbol{\omega}_b - S(\bar{\boldsymbol{\omega}}_b)\delta\phi_b \quad (S17)$$

where $S(\cdot)$ is the skew-symmetric matrix operator. This equation rigorously maps the angular velocity error $\delta\boldsymbol{\omega}_b$ to the evolution of the orientation error on the manifold.

3) *Kinematic Consistency Coupling*: According to the Centroidal Momentum Matrix (CMM) mapping $\mathbf{h}_{com} = \mathbf{A}_b \boldsymbol{\nu}_b + \mathbf{A}_s \dot{\mathbf{q}}_s$, the variation in base velocity $\delta \boldsymbol{\nu}_b = [\delta \mathbf{v}_b^\top, \delta \boldsymbol{\omega}_b^\top]^\top$ is coupled with the momentum and spinal joint velocities. By linearizing this relationship around the nominal trajectory, we obtain:

$$\delta \boldsymbol{\nu}_b = \bar{\mathbf{A}}_b^{-1} (\delta \mathbf{h}_{com} - \bar{\mathbf{A}}_s \delta \dot{\mathbf{q}}_s - \delta \mathbf{A}_b \bar{\boldsymbol{\nu}}_b - \delta \mathbf{A}_s \dot{\bar{\mathbf{q}}}_s) \quad (\text{S18})$$

This term is substituted into the position and attitude error dynamics to ensure that the internal shape modulation ($\delta \dot{\mathbf{q}}_s$) explicitly drives the base reorientation error ($\delta \phi_b$).

C. Discrete-Time Linearized Jacobians ($\mathbf{A}_k$ and $\mathbf{B}_k$)

To formulate the discrete-time Quadratic Programming (QP) problem, we integrate the continuous-time error dynamics over the MPC sampling interval Δt using the explicit Euler method (or Runge-Kutta 4th order for higher precision).

Defining the control input deviation as $\delta \mathbf{u}_k = [\delta \mathbf{f}_{c,k}^\top, \delta \dot{\mathbf{q}}_{s,k}^\top]^\top$, the discrete-time state-space representation is formulated as:

$$\delta \mathbf{x}_{k+1} = \mathbf{A}_k \delta \mathbf{x}_k + \mathbf{B}_k \delta \mathbf{u}_k \quad (\text{S19})$$

The highly structured system matrix $\mathbf{A}_k \in \mathbb{R}^{17 \times 17}$ and input matrix $\mathbf{B}_k \in \mathbb{R}^{17 \times (3n_c + 5)}$ are constructed as block matrices:

$$\mathbf{A}_k = \mathbf{I}_{17 \times 17} + \Delta t \begin{bmatrix} \mathbf{0} & \frac{\partial \dot{\mathbf{h}}}{\partial \mathbf{p}} & \frac{\partial \dot{\mathbf{h}}}{\partial \phi} & \mathbf{0} \\ \bar{\mathbf{A}}_{b,v}^{-1} & \mathbf{0} & \frac{\partial \mathbf{v}_b}{\partial \phi} & \frac{\partial \mathbf{v}_b}{\partial \mathbf{q}_s} \\ \bar{\mathbf{A}}_{b,\omega}^{-1} & \mathbf{0} & -S(\bar{\boldsymbol{\omega}}_k) + \frac{\partial \boldsymbol{\omega}_b}{\partial \phi} & \frac{\partial \boldsymbol{\omega}_b}{\partial \mathbf{q}_s} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (\text{S20})$$

$$\mathbf{B}_k = \Delta t \begin{bmatrix} \mathbf{W}_{1,k} & \cdots & \mathbf{W}_{n_c,k} & \mathbf{0} \\ \mathbf{0} & \cdots & \mathbf{0} & -\bar{\mathbf{A}}_{b,v}^{-1} \bar{\mathbf{A}}_s \\ \mathbf{0} & \cdots & \mathbf{0} & -\bar{\mathbf{A}}_{b,\omega}^{-1} \bar{\mathbf{A}}_s \\ \mathbf{0} & \cdots & \mathbf{0} & \mathbf{I}_{5 \times 5} \end{bmatrix} \quad (\text{S21})$$

where $\bar{\mathbf{A}}_{b,v}^{-1}$ and $\bar{\mathbf{A}}_{b,\omega}^{-1}$ are the corresponding block rows of the inverse base inertia matrix $\bar{\mathbf{A}}_b^{-1}$. The non-zero partial derivatives are analytically evaluated along the nominal trajectory to populate $\mathbf{A}_k$ and $\mathbf{B}_k$ at each control step k , ensuring recursive feasibility and singularity-free optimization on the $SE(3)$ manifold.